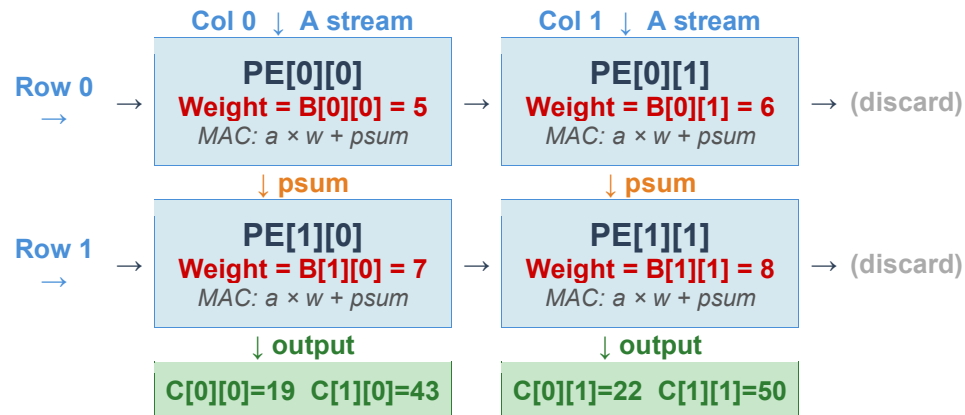


1. PE Diagram with Preloaded Weights

In weight-stationary dataflow, elements of Matrix B are preloaded into PEs and remain fixed. Matrix A elements stream horizontally through rows. Partial sums flow vertically downward.



2. Cycle-by-Cycle Trace Table

Matrix A inputs are staggered so partial sums align correctly as they flow downward. Row 0 receives column 0 of A (1, then 3). Row 1 receives column 1 of A (2, then 4), delayed by 1 cycle.

Cycle	Row 0 In	Row 1 In	PE[0][0] psum	PE[0][1] psum	PE[1][0] psum	PE[1][1] psum	Output C
1	1	0	$1 \times 5 = 5$	$0 \times 6 = 0$	$0 \times 7 + 0 = 0$	$0 \times 8 + 0 = 0$	—
2	3	2	$3 \times 5 = 15$	$1 \times 6 = 6$	$2 \times 7 + 5 = 19$	$0 \times 8 + 0 = 0$	$C[0][0]=19$
3	0	4	$0 \times 5 = 0$	$3 \times 6 = 18$	$4 \times 7 + 15 = 43$	$2 \times 8 + 6 = 22$	$C[1][0]=43$ $C[0][1]=22$
4	0	0	$0 \times 5 = 0$	$0 \times 6 = 0$	$0 \times 7 + 0 = 0$	$4 \times 8 + 18 = 50$	$C[1][1]=50$

Final result: $C = [[19, 22], [43, 50]]$ ✓ matches expected output

3. Data Counts

- (a) Total MAC operations: 2 MACs per output element $\times$ 4 output elements = 8 total MACs.
- (b) Input reuse: Each element of A is reused 2 times (passes through 2 PEs horizontally). Each element of B is reused 2 times (2 different A inputs stream past its fixed weight).
- (c) Off-chip memory accesses:
 - Matrix A (inputs): 4 accesses (streamed in once, staggered across cycles)
 - Matrix B (weights): 4 accesses (loaded once at startup — key advantage of weight-stationary)
 - Matrix C (outputs): 4 accesses (written back once per element)**Total off-chip accesses: 12**

4. Output-Stationary Dataflow

If this were an output-stationary array, the partial sums (elements of output Matrix C) would stay fixed in the PEs to accumulate in place, while elements of both Matrix A and Matrix B stream through them.